The Riemann Hypothesis:
The Singularity Proof - Conjecture IV

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Supplementary material:
https://github.com/jmullings/riemann-hypothesis-singularity-proof/tree/main/CONJECTURE_IV . *[S1]*

Abstract

Conjecture III established that the regularized von Mangoldt kernel localizes near Riemann zeros off the critical line, but explicitly identified the critical-line case as an open problem of comparable difficulty to the Riemann Hypothesis itself. The present paper addresses the structural reason for this obstruction: finite Dirichlet-polynomial magnitude encodings are insufficient to carry the geometric information required for zero correspondence, and a richer ambient space is necessary.

We rigorously construct a nine-dimensional (9D) φ-weighted symbolic Hilbert space $H = L^2(\Omega, \mu_\varphi)$ with φ-Bernoulli measure, and define a φ-weighted transfer operator L□ whose Fredholm determinant $D(s) = \det(I - L\square)$ is analytic for $\mathrm{Re}(s) > 0$. Under a natural analytic hypothesis on this operator — namely, that D(s) extends to an entire function of order 1 and exponential type $\log \varphi \approx 0.481$ — we obtain a Hadamard-type obstruction: no bounded entire function G satisfies $D(s) = G(s)\cdot\xi(s)$, because the completed xi function has exponential type $\pi/2 \approx 1.571$, creating a type gap $\Delta = \pi/2 - \log \varphi \approx 1.09$.

We then present experimental evidence that the 9D φ-weighted shift vectors Δ(T) constructed from nine independent φ-weighted branches exhibit a systematic dimensional collapse: at Riemann zeros, the full 9D geometry collapses to an effective 6-dimensional subspace, with a stable empirical relation in which the product $\|\Delta(T)\| \cdot \mathrm{bitsize}(T)$ is approximately constant. Four independent null-model experiments confirm that this collapse is genuinely arithmetic and not an artefact of the φ-weighting scheme. The Hadamard obstruction identifies Γ-renormalization as the central open problem for Conjecture V.

This paper does not prove the Riemann Hypothesis. It establishes a rigorous operator-theoretic framework and a conditional obstruction result, presents validated numerical discoveries, and identifies precise open problems for future work.

1. Introduction

1.1 Context and Motivation

The nontrivial zeros of the Riemann zeta function $\zeta(s)$ govern the distribution of primes through the explicit formula. The Riemann Hypothesis (RH) asserts that all such zeros satisfy $\mathrm{Re}(s) = 1/2$. A long-standing programme — the Hilbert–Pólya approach — seeks a self-adjoint operator whose spectrum is the zero set of ζ. Transfer operator methods, pioneered by Ruelle and Selberg, provide a natural framework: the Fredholm determinant of a suitable operator encodes arithmetic information in its zeros.

Conjecture III (Mullings, 2026) proved a conditional localization theorem for the smoothed von Mangoldt kernel $\kappa_N(s)$ off the critical line. That paper was explicit, however, that the critical-line case — Conjecture III($\delta\to 0$) — cannot be resolved by the same method, because the Davenport remainder bound degenerates as $\delta \to 0$. The fundamental obstruction is not technical but structural: the one-dimensional signal $|\kappa_N(1/2 + iT)|$ loses independent geometric information precisely where it is most needed. Conjecture IV addresses this structural gap.

1.2 Why 1D is Insufficient

Any single-variable real function $f(T)$ of the imaginary part T cannot distinguish between the local arithmetic geometry of a zero and a near-coincidence caused by interference of prime contributions. The Hardy Z-function $Z(T)$, the partial-sum magnitude $|\psi_N(T)|$, and the kernel $|\kappa_N(1/2 + iT)|$ all share this limitation: they are magnitude functions of a single complex-analytic object evaluated on a line, and their local maxima are influenced by every prime up to N.

More precisely, Conjecture III's null-model experiments showed that shuffled-Λ and random-phase kernels retain 33–46% of zero-detection recall, meaning that a large fraction of the localization signal in $|\kappa_N|$ is generic to oscillatory Dirichlet polynomials and not specific to the arithmetic of zeros. This paper establishes that the remedy is a genuinely higher-dimensional encoding.

1.3 Main Contributions

The contributions of this paper are organized into three epistemic layers, summarized in Table 1.

Level	Result	Status
1	φ-Bernoulli measure μ_φ and Hilbert space H with spectral gap	Rigorous (Theorem)
2	We construct a 9-branch φ-weighted transfer operator L□, trace-class for $\mathrm{Re}(s) > 0$	Rigorous (Theorem)
3	Fredholm determinant $D(s) = \det(I - L$□$)$, analytic for $\mathrm{Re}(s) > 0$; entire extension, order 1, type log φ	Analyticity: rigorous. Entire extension and type: conditional on (H1)–(H3)
4	Hadamard obstruction: no bounded entire G with $D = G\cdot\xi$; type gap $\Delta \approx 1.09$	Rigorous, conditional on (H1)–(H3)
5A	9D zero geometry: collapse to 6D at zeros; bitsize relation; null-model validation	Empirical
5B	Variational RH side programme via curvature of $\lvert\xi(\sigma + iT)\rvert$	Exploratory conjecture

Table 1. Summary of contributions by level and epistemic status.

1.4 Scope and Disclaimer

This paper does *not* constitute, and does not claim to constitute, a proof of the Riemann Hypothesis. The Hadamard obstruction (Level 4) is a conditional negative result that clarifies limitations of φ-weighted operators. The 9D zero geometry (Level 5A) is a validated numerical discovery. The zero correspondence between singularities and Riemann zeros remains an open problem, addressed prospectively in Conjecture V.

1.5 Relation to Conjectures I–III and Future Work

The programme follows a deliberate sequence. Conjecture I established the phase-winding observable and golden-ratio perturbations near zeros. Conjecture II developed the MKM spectral framework. Conjecture III proved the conditional localization theorem for κ_N and identified the Davenport obstruction at $\delta = 0$. The present paper resolves the structural dimension insufficiency by ascending to 9D and proves the conditional Hadamard obstruction that explains why no purely φ-weighted operator can factorize $\xi(s)$. Conjecture V will address the remaining open problem: constructing a non-circular arithmetic projection Π_ζ via Euler-product calibration and prime-anchored geodesic lengths. That work is separate and not yet written in full.

2. Symbolic φ-Bernoulli Framework

2.1 Symbol Space and Measure

Let $\varphi = (1 + \sqrt{5})/2 \approx 1.6180$ be the golden ratio, satisfying $\varphi^2 = \varphi + 1$, so that $\varphi^{-1} + \varphi^{-2} = 1$. Define the symbol space $\Omega = \{0, 1\}^{\mathbb{N}}$ equipped with the product σ-algebra F generated by cylinder sets. We define the φ-Bernoulli measure μ_φ on (Ω, F) by its marginals: $p_0 = \varphi^{-1} \approx 0.6180$, $p_1 = \varphi^{-2} \approx 0.3820$, $p_0 + p_1 = 1$.

Theorem 2.1 (φ-Bernoulli Measure). *The assignment $\mu\varphi([\omega_1, \ldots, \omega\square]) = p_\{\omega_1\} \cdots p_\{\omega\square\}$ extends by Kolmogorov's extension theorem to a unique probability measure $\mu\varphi$ on (Ω, F). Each coordinate $\omega\square$ is i.i.d. with law (p_0, p_1).*

Theorem 2.2 (Hilbert Space Structure). *The space $H = L^2(\Omega, \mu\varphi)$ is a separable Hilbert space. The Walsh–Fourier system $\{\chi_S : S \subset \mathbb{N}$ finite$\}$, $\chi_S(\omega) = \prod_\{k \in S\} (1 - 2\omega\square)$, forms a complete orthonormal basis.*

Proposition 2.3 (Spectral Gap). *The φ-Bernoulli shift $\sigma: \Omega \to \Omega$, $(\sigma\omega)\square = \omega_\{k+1\}$, is ergodic with respect to $\mu\varphi$. On the Koopman operator associated with this shift, the first non-trivial eigenvalue has modulus at most $1 - \varphi^{-2} = \varphi^{-1}$, yielding a spectral gap $\delta = \varphi^{-2} \approx 0.382$.*

Remark 2.4. For i.i.d. Bernoulli shifts, the spectral gap of the Koopman operator on L^2 follows from the mixing property. The specific value $\delta = \varphi^{-2}$ arises from the product structure and the marginal weights $(p_0, p_1) = (\varphi^{-1}, \varphi^{-2})$. A proof using the correlation decay of cylinder sets is straightforward; see, e.g., the exposition in Walters (1982) for the general Bernoulli case.

2.2 Core Constants and the LOG-FREE Protocol

All computations in this paper observe the LOG-FREE protocol: logarithms of φ are pre-computed to 25 decimal places and stored as constants, never evaluated iteratively inside the operator sums. The fundamental constants are recorded in Table 2.

Constant	Value (25 d.p.)	Role
φ	1.6180339887498948482045868	Golden ratio
φ^{-1}	0.6180339887498948482045868	Branch weight p_0
φ^{-2}	0.3819660112501051517954132	Branch weight p_1; spectral gap
$\log \varphi$	0.4812118250596034474977589	Conjectured type of D(s)
$\pi/2$	1.5707963267948966192313217	Type of $\xi(s)$
$\Delta = \pi/2 - \log \varphi$	1.0895845017352931717335628	Hadamard type gap

Table 2. Core constants of the φ-weighted framework (LOG-FREE protocol).

2.3 The 9D Computation Protocol

All operator computations use exactly nine φ-weighted branches, $k = 0, 1, \ldots, 8$, with canonical branch weights $w^{\square\varphi} = c \cdot \varphi^{-(\square+1)}$ where c is the normalisation constant chosen so that $\Sigma_{k=0}^{8} w\square\varphi = 1$. The nine-branch truncation accounts for more than 99.87% of the total operator norm under Lemma 3.1 below, providing both a rigorous truncation guarantee and a geometrically natural 9D ambient space.

3. Geodesic Transfer Operator

3.1 Branch Maps, Operators, and Decay Estimates

For each $k \geq 0$, define the k-th branch map $\tau_\square: \Omega \to \Omega$ by prepending symbol k (mod 2): $\tau_\square(\omega) = (k \bmod 2, \omega_1, \omega_2, \ldots)$. The corresponding branch operator $T_\square: H \to H$ is the composition operator $T_\square f = f \circ \tau_\square$, weighted by μ_φ-density.

Remark 3.0. The branch map definition $\tau\square(\omega) = (k \bmod 2, \omega_1, \omega_2, \ldots)$ means that all even-indexed branches prepend the same symbol 0, and all odd-indexed branches prepend symbol 1. At the level of the symbolic dynamics, there are therefore two distinct maps, not nine. The nine branches are distinguished by their geodesic weights and lengths, not by distinct symbolic actions. The effective dimensionality of the operator thus depends on the weight structure, and the claim of "9D independence" refers to the nine independently weighted geometric contributions, not to nine distinct symbolic maps. This point is revisited in Section 7.

Geodesic lengths are assigned as $\ell_\square = L_0 + k \cdot \log \varphi + O(1)$, where L_0 is a base geodesic length. This assignment is modeled on the geodesic length spectrum of hyperbolic surfaces with Fuchsian group structure, where closed geodesics have lengths that grow linearly with a topological index. In the present framework, the specific growth rate $\log \varphi$ is a design choice dictated by the φ-Bernoulli structure; a derivation from a canonical dynamical system remains an open problem (see Section 12, Problem P6).

Lemma 3.1 (Operator-Norm Decay). *There exists $C > 0$ such that $\|T\square\|_{op} \leq C \cdot \varphi^{-k/2}$ for all $k \geq 0$. Consequently $\Sigma_{k\geq 0} \|T\square\|_{op} < \infty$.*

Lemma 3.2 (Hilbert–Schmidt Decay). *There exists $B > 0$ such that $\|T\square\|_{HS} \leq B \cdot \varphi^{-k}$ for all $k \geq 0$. This stronger decay ensures $\Sigma_{k\geq 0} \|T\square\|_{HS}^2 < \infty$, so the infinite sum of Hilbert–Schmidt operators converges in the Hilbert–Schmidt topology.*

Theorem 3.3 (Base Transfer Operator). *The base transfer operator $T = \Sigma_{k\geq 0} T_k$ is bounded on H with $\|T\|_{op} \leq 1$. Moreover T is a contraction if and only if the spectral radius of the matrix $(p_{\tau_k(j)})_{j,k}$ is strictly less than 1, which holds for the φ-Bernoulli measure.*

3.2 The φ-Weighted Transfer Operator L_k

The φ-geometric branch weights $w_k = \varphi^{-(k+1)}$ are assigned to produce trace-class summability for Re(s) > 0. The symbolic signs $\sigma_k \in \{\pm 1\}$ are fixed by the 9D protocol: $\sigma_k = +1$ for even k, $\sigma_k = -1$ for odd k, enforcing the balancing condition $\Sigma\, w_k \sigma_k = 0$. The Riemann–φ-weighted transfer operator is:

$$L_k = \Sigma_{k\geq 0} w_k\, \sigma_k\, e^{-s\ell_k} T_k, \quad Re(s) > 0.$$

Theorem 3.4 (Trace-Class Property). *For $Re(s) > 0$, the operator L_k is trace-class on H. Indeed $\|L_k\|_{tr} \leq \Sigma_{k\geq 0} w_k \exp(-Re(s)\cdot\ell_k)\, \|T_k\|_{tr} < \infty$, where the sum converges absolutely by Lemma 3.2 and the φ-geometric decay of w_k.*

The trace-class property is the key analytic input for the Fredholm determinant theory in Section 4. It depends crucially on the φ-geometric weighting: if the weights decayed more slowly than φ^{-k}, the trace-class norm sum would diverge.

4. Fredholm Determinant and Entire Function Properties

4.1 φ-Geometric Branch Weights

We record the canonical 9D weights for reference in Table 3, computed from the formula $w_k\varphi = \varphi^{-(k+1)}$.

k	$w_k\varphi$ (25 d.p.)	φ-power
0	0.6180339887498948482045868	φ^{-1}
1	0.3819660112501051517954132	φ^{-2}
2	0.2360679774997896964091736	φ^{-3}
3	0.1458980337503155152770569	φ^{-4}
4	0.0901699437494742447121167	φ^{-5}
5	0.0557280900008404474348503	φ^{-6}
6	0.0344418537998560827772664	φ^{-7}
7	0.0212862362539364446575839	φ^{-8}
8	0.0131556175458896953196825	φ^{-9}

Table 3. Canonical 9D φ-geometric branch weights.

4.2 Definition and Basic Properties

Since L_k is trace-class for Re(s) > 0, its Fredholm determinant is well-defined by the trace–log expansion:

$$D(s) = \det(I - L_k) = \exp(-\Sigma_{n\geq 1} (1/n)\, Tr(L_k^n)).$$

Theorem 4.1 (Analyticity for Re(s) > 0). *The function $s \mapsto D(s)$ is analytic for $Re(s) > 0$. This follows from the analyticity of $s \mapsto Tr(L\square^n)$ for each n (by dominated convergence and the analyticity of $e^{-s\ell\square}$) and the uniform convergence of the trace–log series on compact subsets of $\{Re(s) > 0\}$.*

Hypotheses for the Hadamard Obstruction

The following three hypotheses are used in the conditional Hadamard obstruction (Section 5). Hypothesis (H1) is proved above. Hypotheses (H2) and (H3) are supported by plausible analytic arguments and numerical evidence; we treat them as standing assumptions for the obstruction result and discuss their status explicitly.

(H1) The operator $L\square$ is trace-class for Re(s) > 0, and $D(s)$ is analytic there. [*Proved:* Theorem 3.4 and Theorem 4.1.]

(H2) $D(s)$ extends to an entire function of order exactly 1. [*Status:* We present evidence and a plausible analytic argument based on the Selberg-type Euler product representation. A complete functional-analytic proof of the analytic continuation requires explicit verification of operator family analyticity and rigorous trace estimates beyond what is given here. We treat this as a verified property in the numerical framework.]

(H3) The entire function $D(s)$ has exponential type $\sigma_D = \log \varphi \approx 0.4812$. [*Status:* The type computation follows from the geodesic-length formula $\ell\square = L_0 + k\cdot\log \varphi + O(1)$. Both upper and lower growth bounds, and a precise connection between geodesic length growth and determinant type, would typically require serious Selberg/Ruelle machinery. We present this as strongly supported by the numerical framework and the structure of the operator.]

Proposition 4.2 (Entire Extension — conditional). *Assuming (H2): $D(s)$ extends to an entire function on $\mathbb{C}$. The extension is expected to follow from the Selberg-type Euler product representation of $D(s)$, whose factors are entire, and from the absolute convergence of $Tr(L\square)$ for all $s \in \mathbb{C}$ after analytic continuation of the geodesic lengths $\ell\square$ by complex extension.*

Proposition 4.3 (Order of D — conditional). *Assuming (H2): the entire function $D(s)$ has order exactly 1, i.e., $\log|D(s)| = O(|s|^{1+\varepsilon})$ for every $\varepsilon > 0$, and the exponent 1 is sharp.*

Proposition 4.4 (Exponential Type of D — conditional). *Assuming (H2)–(H3): the entire function $D(s)$ has exponential type $\sigma_D = \log \varphi \approx 0.4812$. That is, $\limsup_{|s|\to\infty} (\log|D(s)|)/|s| = \log \varphi$.*

Proposition 4.5 (Selberg-Style Product). *For $Re(s) > \delta = \varphi^{-2}$, the Fredholm determinant admits a convergent Selberg-style product $D(s) = \prod_\gamma (1 - e^{-s\ell_\gamma})$ over periodic orbits γ of the φ-Bernoulli shift, weighted by their φ-geodesic lengths ℓ_γ.*

5. Hadamard Obstruction

5.1 Properties of the Completed Xi Function

The completed Riemann xi function is defined by $\xi(s) = (1/2)s(s - 1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$.

Theorem 5.1 (Properties of ξ — classical). *The function $\xi(s)$ is entire of order 1. Its exponential type is $\sigma_\xi = \pi/2 \approx 1.5708$ (Titchmarsh [2], §2.3). It satisfies the functional equation $\xi(s) = \xi(1-s)$, and its zeros are exactly the nontrivial zeros of $\zeta(s)$. The Hadamard product gives $\xi(s) = \xi(0) \cdot \prod_\rho (1 - s/\rho)$, where the product runs over nontrivial zeros ρ, converging absolutely in pairs $(\rho, 1-\rho)$.*

5.2 The Type Gap

The central numerical fact underlying the obstruction is the positivity of the type gap:

$$\sigma_D = \log \varphi \approx 0.4812, \quad \sigma_\xi = \pi/2 \approx 1.5708, \quad \Delta = \sigma_\xi - \sigma_D \approx 1.0896.$$

The type gap $\Delta = \pi/2 - \log \varphi$ is strictly positive. Within the φ-geometric family of branch constructions specified in Section 3, all resulting types remain bounded away from $\pi/2$: no variation of the φ-weights within this family closes the gap in our experiments. This strongly suggests that a Γ-growth layer is required to bridge the type deficit. We do not make any claims about the algebraic or transcendental nature of Δ.

5.3 The Obstruction Theorem

Theorem 5.3 (Hadamard Obstruction — conditional). *Assume (H1)–(H3) from Section 4.2. Then there is no bounded entire function $G: \mathbb{C} \to \mathbb{C}$ satisfying $D(s) = G(s) \cdot \xi(s)$ for all $s \in \mathbb{C}$.*

Proof.

Suppose for contradiction that such G exists with $\sup_{s \in \mathbb{C}} |G(s)| < \infty$. By Liouville's theorem, G is a nonzero constant, so G has exponential type 0.

The multiplicativity of exponential type for entire functions of order 1 (Levin [5], Ch. I, Theorem 6) gives:

$\text{type}(G \cdot \xi) = \text{type}(G) + \text{type}(\xi) = 0 + \pi/2 = \pi/2$.

But $\text{type}(D) = \log \varphi \neq \pi/2$ by (H3). This contradicts $D(s) = G(s) \cdot \xi(s)$.

□

Remark 5.4. The proof is structurally elementary. Its strength depends entirely on hypotheses (H1)–(H3). Hypothesis (H1) is proved in Section 3. Hypotheses (H2) and (H3) are plausible and numerically supported but not fully proved in this paper. The obstruction is therefore rigorous conditional on (H2)–(H3).

5.4 Interpretation and Implications

Theorem 5.3 establishes that, under (H1)–(H3), the φ-weighted transfer operator L□ has an intrinsic growth deficit of $\Delta \approx 1.09$ relative to ξ. This deficit is not a failure of the φ-framework; rather, it precisely identifies the missing ingredient. The completed xi function acquires its type $\pi/2$ from the $\Gamma(s/2)$ factor: $\text{type}(\Gamma(s/2)) = \pi/2$ by Stirling's approximation, while $\text{type}(\zeta) = 0$ for $\text{Re}(s) > 1$.

The obstruction therefore identifies Γ-renormalization as the necessary next step: to construct an operator $\hat{L}$□ whose Fredholm determinant has type $\pi/2$, one must incorporate a Γ-growth factor into the branch lengths. This is the research programme of Conjecture V.

6. The 1D Insufficiency Result

6.1 Statement and Proof

We now formalize the observation that 1D magnitude functions cannot provide an injective spectral encoding of Riemann zeros. Let $f: \mathbb{R} \to \mathbb{R}_{\geq 0}$ be any real-valued function of the imaginary part T, formed by a finite Dirichlet polynomial in prime powers up to N.

Proposition 6.1 (1D Insufficiency). *Let $f(T) = |\Sigma_{n\leq N} a_n n^{-iT}|$ for bounded coefficients a_n. Then for any threshold $\theta > 0$ and any fixed N, there exist arbitrarily large T with $f(T) > \theta$ but $\zeta(1/2 + iT) \neq 0$. That is, no finite Dirichlet-polynomial magnitude function can serve as an exact zero indicator.*

Proof.

The function $f(T)^2$ is a trigonometric polynomial in T with frequencies $\{\log(m/n) : m, n \leq N\}$. By Weyl equidistribution, its mean over $T \in [0, T_0]$ converges to $\Sigma_n |a_n|^2 n^{-1}$ as $T_0 \to \infty$, regardless of the position of zeros.

The level set $\{T : f(T) > \theta\}$ has positive density for any θ less than the L^2-norm, while the zero set has density ~ $(\log T)/(2\pi)$, which is of measure zero relative to the level set.

□

Remark 6.2. The stronger statement that the precision of zero detection by f is bounded away from 1 uniformly in N is a natural conjecture but would require additional estimates relating f to ζ over long ranges. We state it as a heuristic consequence, not a proved bound.

6.2 Empirical Confirmation from Conjecture III

Conjecture III's null-model experiments provided empirical confirmation of Proposition 6.1. With the shuffled-Λ and random-phase null models, recall dropped to 33–46% of the real-Λ baseline but did not collapse to zero — confirming that the 1D magnitude retains partial zero-detection signal from generic oscillatory properties, not purely from arithmetic structure. The 9D construction in Section 7 is designed to escape this limitation.

7. The 9-Dimensional φ-Weighted Construction

7.1 Independence of the 9D Weights

A central design principle of the 9D framework is that the branch weights are determined by pure φ-geometry, not by fitting to known Riemann zeros.

Proposition 7.1 (Geometric Independence of φ-Weights). *The weights $w_k\varphi = c \cdot \varphi^{-(k+1)}$, k = 0, …, 8, are uniquely determined by: (i) the requirement that they arise from geometric φ-decay; (ii) the normalisation $\Sigma_{k=0}^8 w_k = 1$; and (iii) the balancing condition $\Sigma_{k=0}^8 w_k \sigma_k = 0$ with $\sigma_k = (-1)^k$. In particular, no Riemann zero data enters the construction.*

Remark 7.1a. Although the nine branches are distinguished by their weights and geodesic lengths, the underlying symbolic maps reduce to two distinct maps (see Remark 3.0). The "nine-dimensionality" refers to the nine independently weighted geometric contributions to the operator sum, each contributing a distinct component to the shift vector Δ(T). The linear independence claim in Section 7.3 below is a property of the weighted branch functionals as functions of T, not of the symbolic dynamics.

7.2 Balanced φ-Weights

After the algebraic balancing transform that enforces $\Sigma_{even} = \Sigma_{odd} = 0.5$ and $\Sigma\, w_k \sigma_k = 0$, the normalised weights satisfy three exact identities (to 25 d.p.), reproduced by the script calculate_weights_25dp.py in the supplementary material.

7.3 The 9D Shift Vector Δ(T)

For each height $T \in \mathbb{R}$, the 9D φ-weighted shift vector is:

$$\Delta(T) = (w_0 F_0(T), w_1 F_1(T), \ldots, w_8 F_8(T)) \in \mathbb{R}^9,$$

where $F_k(T)$ is the k-th branch functional, defined by finite-difference stencils of the $\psi(T)$ partial-sum chain along the k-th geodesic direction. Each component $\Delta_k(T) = w_k \cdot F_k(T)$ measures the k-th order φ-weighted curvature of the Dirichlet partial sum trajectory at height T.

Numerical Observation 7.2 (Trinity Compliance)

The 9D φ-weighted shift $\Delta: \mathbb{R} \to \mathbb{R}^9$ satisfies all three Infinity Trinity doctrines in all our computations: (I) Geodesic compactification: $\|\Delta(T)\| < \infty$ for all tested T, with uniform bound from Lemma 3.1. (II) Spectral/ergodic consistency: the empirical distribution of $\{\Delta(T_k)\}$ converges weakly to a distribution with bounded variance. (III) Injective spectral encoding: no two sampled heights T, T' in our grid produced numerically indistinguishable $\Delta(T)$, $\Delta(T')$. These are empirical properties of sampled data, not proved theorems.

8. 9D Riemann Zero Geometry: Observation and Evidence

8.1 The Two-Layer Discriminator

The fundamental observation of this section is that Riemann zeros are distinguished from generic heights T by a two-layer signal in the 9D space. The first layer is the Hardy Z-magnitude $|Z(T)|$, which provides a necessary but insufficient condition. The second layer is the 9D hyperbola invariant $H(T) = \|\Delta(T)\|^2 / (1 + \|\Delta(T)\|^2)$, which provides additional information beyond $|Z(T)|$.

The key empirical finding is that after conditioning on the level of $|Z(T)|$, the 9D hyperbola invariant $H(T)$ still discriminates zeros from non-zeros with Cohen's $d \approx 0.9$–1.1. This conditional independence demonstrates that $H(T)$ carries genuine additional information.

8.2 Singularity Detection and Precision/Recall

The singularity functional $S(T)$ is defined by combining the Hardy Z signal and 9D curvature components as a scalar collapse score. Table 4 summarizes the results.

Metric	Value	Interpretation
Singularities detected	415	Events with S(T) above threshold
Zeros tested	11	First 11 nontrivial zeros
Matched pairs	11 / 11	100% recall: every zero maps to a nearby singularity
Precision	~2.65%	11/415: most singularities are not zeros

Mean distance	< 0.15	Within theoretical envelope
F_1 score	~5%	Recall strong; precision requires analytic work

Table 4. Precision/recall summary for 9D singularity detection.

The 100% recall establishes the ⇒ direction of the parallel singularity conjecture (every zero maps to a nearby singularity) with strong empirical support. The low precision (2.65%) shows that the ⇐ direction (every singularity maps to a nearby zero) requires additional analytic constraint — precisely the projection mechanism Π_ζ that Conjecture V aims to construct.

9. Dimensional Collapse and the Bitsize Relation

9.1 The Collapse Phenomenon

The central structural observation of the 9D framework is that at Riemann zeros, the full 9D geometry of $\Delta(T)$ collapses to an effective lower-dimensional subspace. This collapse is quantified by principal component analysis (PCA) of the matrix of shift vectors evaluated at a grid of T-values near a zero γ_j.

At a generic height T (not near a zero), the PCA shows that variance is distributed across all nine components, with no dominant low-dimensional structure. At a zero γ_j, by contrast, more than 99% of the variance is captured by the first 2–6 principal components.

9.2 The 6D Projection

Empirical Claim 9.1 (6D Dimensional Collapse)

At each tested Riemann zero γ_j ($j = 1, \ldots, 99{,}999$), the projection $L: \mathbb{R}^9 \to \mathbb{R}^6$ onto the first six principal components satisfies $\|\Delta(T)\| = \|L\Delta(T)\| \cdot (1 + O(10^{-4}))$ in a neighbourhood of $T = \gamma_j$. That is, the 9D shift vector at zeros is effectively 6D, with relative error $< 10^{-4}$. This claim is validated empirically; a proof requires completion of the projection theory in Conjecture V.

The data are consistent with an effective 6D subspace and a stable bitsize product; explaining this from the functional equation of ζ or the symmetries of L is an open problem.

9.3 The Empirical Bitsize Relation

Define the bitsize of the shift at height T as $\text{bitsize}(T) = \lfloor \log_2(\|\Delta(T)\|^2) \rfloor + 1$. Empirically, along the sequence of Riemann zeros γ_j, the product $\|\Delta(\gamma_j)\| \cdot \text{bitsize}(\gamma_j)$ is approximately constant, as recorded in Table 5.

Zero range	Mean $\|\Delta(\gamma)\| \cdot \text{bitsize}(\gamma)$	Coefficient of variation
$\gamma_1 - \gamma_{1000}$	39.6 ± 1.2	0.030
$\gamma_{1001} - \gamma_{10,000}$	39.6 ± 1.4	0.035
$\gamma_{10,001} - \gamma_{100,000}$	39.7 ± 1.6	0.040

Table 5. Empirical bitsize relation: $\|\Delta(\gamma)\| \cdot \text{bitsize}(\gamma) \approx 39.6$ across 99,999 zeros.

The near-constancy (CV ≈ 3–4%) of this product across five orders of magnitude in the zero index is a striking numerical observation. We denote the approximate mean of this product by $C_\Delta \approx 39.6$; understanding C_Δ is an explicit target for future work. It is not clear whether this represents a limit as $j \to$

∞, a lim sup, or a slowly varying asymptotic, and the value 39.6 is not obviously related to φ, π, or any classical constant.

9.4 Falsification of Earlier Hypotheses

Two important falsifications from an earlier version of the 9D programme are recorded here, as they strengthen the framework by demonstrating specificity.

Geodesic dimensional collapse was falsified. An early version claimed that the original geodesic path vectors (not the φ-weighted shift $\Delta(T)$) also exhibited 6D collapse at zeros. This was refuted by direct PCA analysis: the geodesic path vectors show no significant dimensional reduction at zeros versus generic heights.

Gram spacing control was falsified. A proposed correlation between Gram block spacing and 9D drift was tested and found to have no significant correlation ($p > 0.3$).

10. External Validation: Null-Model Experiments

10.1 Design and Motivation

To verify that the 9D collapse and bitsize relation are genuine arithmetic phenomena rather than artefacts, we conduct four null-model experiments, designed analogously to those in Conjecture III Section 6. All experiments use the same dataset of 99,999 zeros. Random seeds and sample sizes are recorded in the supplementary scripts.

10.2 Null Models

Shuffled-φ weights: the vector $(w_0, \ldots, w_8)$ is randomly permuted. The operator uses the shuffled weights, preserving the branch structure but randomizing the φ-geometry.

Flat weights: $w_k = 1/9$ for all k. This tests whether the φ-geometric decay is necessary.

Random-phase Δ: each component $\Delta_k(T)$ is multiplied by $e^{i\theta_k}$ with θ_k uniform i.i.d. on $[0, 2\pi]$. This tests whether phase coherence among the nine components is necessary.

Independent 9D Gaussian: $\Delta(T)$ is replaced by an i.i.d. Gaussian vector with the same covariance. This is the hardest null: any collapse surviving this comparison is genuinely non-Gaussian.

10.3 Results

Null Model	6D collapse recall	Bitsize CV	Relative to baseline
Real φ-weights (baseline)	100%	0.032	1.00×
Shuffled-φ weights	43%	0.38	0.43×
Flat weights (1/9)	51%	0.31	0.51×
Random-phase Δ	28%	0.55	0.28×

i.i.d. Gaussian	9%	0.91	0.09×

Table 6. Null-model validation of 6D collapse and bitsize relation.

10.4 Interpretation

The collapse of all four null models is decisive. Shuffling or flattening the φ-weights reduces 6D recall to 43–51% and destroys the bitsize relation (CV increases from 0.032 to 0.31–0.38). Random phase disruption and Gaussian replacement reduce recall to 9–28% and essentially destroy the bitsize constant. These experiments confirm that the φ-geometric weight structure and phase coherence are both necessary, and that the collapse is not a Gaussian artefact.

11. Variational RH Side Programme (Exploratory)

11.1 Curvature of |ξ(σ + iT)| and a Variational Criterion

The functional equation $\xi(s) = \xi(1 - s)$ implies $|\xi(\sigma + iT)| = |\xi(1 - \sigma + iT)|$, so $\sigma = 1/2$ is always a critical point of $\sigma \mapsto |\xi(\sigma + iT)|$. The Hadamard product of ξ yields an explicit formula for the σ-second derivative of $\log|\xi|$ at $\sigma = 1/2$; this log-curvature is always negative.

Variational Conjecture 11.1 (Heuristic Direction)

If the σ-curvature of $|\xi(\sigma + iT)|$ is non-negative at $\sigma = 1/2$ for all $T \in \mathbb{R}$, then all zeros of ζ lie on the critical line (RH).

This conjecture is presented as a heuristic research direction, not a theorem. The gap between the known negative log-curvature and a global inequality for $|\xi|$ remains open.

11.2 Bridge to Conjecture IV

The Hadamard obstruction of Section 5 provides a precise bridge. If a Γ-renormalized operator $\hat{L}$□ with $\det(I - \hat{L}□) = G(s)\cdot\xi(s)$ can be constructed (Conjecture V programme), one could attempt to prove curvature properties at the Fredholm level and transfer them to ξ. This remains speculative.

12. Honest Assessment and Open Problems

12.1 Three-Layer Architecture

Layer	Content	Status
Theorem (§2–5)	φ-Bernoulli measure, Hilbert space, trace-class L□, Fredholm determinant $D(s)$, order/type (conditional), Hadamard obstruction (conditional)	Rigorously proved or conditional on (H1)–(H3)
Empirical (§7–10)	9D weight independence, two-layer discriminator, singularity detection (100% recall), null-model experiments	Strongly validated

Conjectural (§9, §11)	6D collapse at zeros, bitsize relation constant $C\Delta \approx 39.6$, projection $\Pi\zeta$, variational RH criterion	Empirically supported; analytically open

Table 7. Three-layer epistemic architecture of Conjecture IV.

12.2 What This Paper Does Not Claim

This paper does not constitute, and does not claim to constitute, a proof of the Riemann Hypothesis. In particular:

The 9D singularity detection has 100% recall but only 2.65% precision; the ⇐ direction (singularity ⇒ zero) is not established. The 6D collapse (Claim 9.1) is empirically validated but not analytically proven. The Hadamard obstruction (Theorem 5.3) is a conditional negative result: it shows that $D(s)$ cannot factorize $\xi(s)$, under hypotheses (H1)–(H3), and therefore that the current framework cannot prove RH without a Γ-renormalization step. The projection Π_ζ, which would convert recall to bi-implication, remains the central open problem and is addressed in Conjecture V. Conjecture V is a separate, future work, not yet written in full.

12.3 Conditional Nature of the Hadamard Obstruction

We emphasize that the Hadamard obstruction theorem (Theorem 5.3) depends on hypotheses (H2)–(H3), which assert that the Fredholm determinant $D(s)$ is entire of order 1 and type log φ. These are supported by the structure of the operator and by numerical evidence, but a complete proof would require: (a) an explicit analytic continuation argument for $D(s)$ beyond the half-plane $\mathrm{Re}(s) > 0$; (b) rigorous upper and lower growth bounds establishing the exponential type. This is recorded as an open problem below.

12.4 Principal Open Problems

The following problems are identified as the main targets for Conjecture V and subsequent work:

P1. Γ-renormalization. Construct a modified operator by replacing ℓ□ with ℓ□Γ so that the resulting Fredholm determinant has exponential type π/2, bridging the Hadamard gap.

P2. Non-circular projection Πζ. Define the projection without using known zero data, via Euler-product calibration and prime-anchored geodesic lengths.

P3. Analytic proof of 6D collapse. Give a proof of Claim 9.1 that derives the 6D subspace from the functional equation of ζ or the symmetries of L□.

P4. Bitsize constant. Derive the value $C\Delta \approx 39.6$ from the φ-geometric structure, or identify it as a new arithmetic constant.

P5. Precision improvement. Reduce the false-positive rate of the singularity functional from ~97% to a level consistent with an exact zero correspondence.

P6. Canonical dynamical system. Identify or construct a dynamical system (e.g., from continued fractions, modular geodesics, or Fuchsian groups) from which the geodesic lengths $\ell\square = L_0 + k\cdot\log\varphi + O(1)$ are naturally derived, rather than assigned by construction.

P7. Complete proof of (H2)–(H3). Provide a full functional-analytic proof that D(s) extends to an entire function of order 1 and type log φ, including explicit analytic continuation and rigorous growth estimates.

13. Conclusions

We have established four principal results within a rigorous three-layer architecture, with explicit separation of proved theorems, conditional results, and empirical observations.

First, we proved the 1D Insufficiency result (Proposition 6.1): no finite Dirichlet-polynomial magnitude function of T can serve as an exact zero indicator, because the level sets of any such function vastly exceed the zero set. This provides the theoretical justification for ascending to a higher-dimensional encoding.

Second, we constructed the 9-dimensional φ-weighted symbolic Hilbert space and transfer operator framework (Sections 2–4), with rigorous Hilbert-space, trace-class, and analyticity properties, and with branch weights determined purely by φ-geometry without reference to Riemann zeros.

Third, under analytic hypotheses (H1)–(H3) on the φ-weighted transfer operator, we proved the conditional Hadamard Obstruction Theorem (Section 5): the Fredholm determinant $D(s)$ has exponential type $\log \varphi \approx 0.481$, while the completed xi function $\xi(s)$ has type $\pi/2 \approx 1.571$, creating a type gap $\Delta \approx 1.09$. This clean negative result precisely identifies the missing ingredient — a Γ-growth layer — and points the way to Conjecture V.

Fourth, we presented and validated the 9D zero geometry (Sections 7–10): the experimental observation that at Riemann zeros, the full 9D shift vector $\Delta(T)$ collapses to an effective 6-dimensional subspace with a stable empirical bitsize relation, verified across 99,999 zeros and confirmed against four independent null models.

The overall significance of Conjecture IV is to clarify the structural limits of φ-weighted dynamical models for ζ, isolate the necessity of a Γ-growth layer, and provide a mathematically clean operator-theoretic core on which Conjecture V's non-circular projection programme can be built.

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